Payton Pei

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Summary

Systems Science and Engineering master's student at the University of Ottawa, graduating in December 2026 (expected). Software engineer building and fine-tuning AI systems with Python, PyTorch, Rust/Tauri and TypeScript/React; published LifeMint on the App Store. Former Nokia and Ford intern seeking new-grad AI application and developer-tools roles in Ottawa.

Education

University of Ottawa
Master of Systems Science and Engineering — *2024 — Dec 2026 (expected)*
Carleton University
Bachelor of Computer Science — *2017 — 2023*

Projects

Réussir en français — french.paytonpei.top
- Built and deployed a TCF Canada exam-prep platform at french.paytonpei.top: Next.js 16 / React 19 frontend with same-origin API rewrites to a Node/SQLite study backend, also runnable on Cloudflare Worker/D1.
- Implemented timed four-section mock exams with compare-and-swap session sync, a 26-week SM-2 study plan, bilingual dictionary, sentence-aligned listening audio, and Stripe billing on a Docker self-hosted VPS.

GaokaoTutor — LLM Fine-Tuning & Evaluation
- Fine-tuned Spark X2.5-4B with 4-bit QLoRA on an RTX 4090 using 161 tutoring traces; built reproducible data provenance, anti-contamination checks and a matched evaluation on a frozen 80-case unseen holdout.
- Reduced unsolicited solution dumping in hint/diagnosis cases from 100% to 13.5% and truncation from 57.5% to 0%; integrated the adapter into a Python tutor service and Expo prototype with deterministic math verification.

pipi-shrimp-agent — github.com/mammut001/pipi-shrimp-agent
- Built a local-first desktop AI agent with Rust/Tauri, React/TypeScript and SQLite, integrating multiple LLM providers, structured tool calls and project memory.
- Implemented provider adapters, buffered SSE parsing and cancellation, plus CDP browser automation with failure recovery and context compression for long-running sessions.

Work Experience

Teaching Assistant — 2021
Carleton University — *Ottawa, Canada*
- Teaching assistant for Python 3 (Jan–Apr 2021) and JOUR 4401 (Sep–Dec 2021), covering relational databases, web crawling and data analysis.

Software Tester Intern — Sep 2020 — Aug 2021
Nokia Canada — *Ottawa, Canada*
- Tested the legacy NFM-P client in the 7750 Service Router context using JavaScript, XML and the Mocha testing framework.
- Documented system architecture and data flows in Confluence.

Software Engineer Intern — May 2020 — Aug 2020
Ford Motor Canada — *Ottawa, Canada*
- Contributed to in-vehicle and cloud connectivity software in C++11 on QNX/Linux.
- Wrote GoogleTest unit tests, fixed bugs and maintained technical documentation; used Jenkins in the CI/CD workflow.

Research

Low-Light RAW Image Restoration with Bridge U-Net and DDBM — 2026
- Responsible for the initial DDBM implementation in a collaborative low-light RAW image restoration study using SID and ELD data.
- The project evaluated a DDBM-Feature variant at 26.12 dB PSNR and 0.807 MS-SSIM on 598 SID Sony test images.

Skills

- **Languages**: Python, TypeScript, JavaScript, Rust, Swift, C++
- **Frameworks**: React, Next.js, Node.js, Tauri 2, SwiftUI
- **AI & Systems**: PyTorch, Transformers, PEFT/TRL, QLoRA, MLX, LLM APIs, Tool Calling, MCP, SQLite
- **Testing & Delivery**: PyTest, Vitest, Playwright, GoogleTest, Jenkins, Docker
- **Additional**: CDP Browser Automation, WatchConnectivity, StoreKit 2